

Tony Siu

FOUNDING AI ENGINEER · MULTI MODAL GENAI RESEARCHER · APPLIED STATISTICS RESEARCHER

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Education

Masters, Computational Data Science | 4.0 of 4.0 GPA

Temple University

Philadelphia, PA

Jan 2024 - On Hold

- IEEE Cloud & AI Student Ambassador

Bachelors, Computational Data Science | 3.87 of 4.0 GPA

Temple University

Philadelphia, PA

Aug 2022 - Dec 2024

- Graduated with Distinction
- Dean's list Fall 2024
- Visual Question Answering Preference Optimization Research under Dr Longin Jan Latecki

Engineering Experience

Founding AI Engineer & Researcher

A16z Speedrun SR005

San Francisco, CA

Apr 2025 - Dec 2025

- Principal Engineer – 60% singlehanded engineering output in a company of 20 people
- 200x optimization – 0.2 sec per 78+ VS 5 sec per 1 animation rendering for spritesheets, .gifs
- Architect & built – Natural language to full GenAI game generation MVP in under a month
- Develop Vision based Promptless UI action Agentic systems for long-context inter-session user profiles.
- Reimplement Bleeding Edge papers such as the Agentic Context Engineering(ACE) paper or Tongyi labs AgentFold paper as CUAs Scaffolds

Generative AI Engineer

Philadelphia, PA

A Pro Inc Solutions

Feb 2025 - July 2025

- Architect end to end systems to generate AI Influencers, Key Opinion Leaders for AI Digital Marketing as a service.
- Design application evaluation criterias to determine the appropriateness of the Generated Videos, Audio, Images of the AI Influencer.
- 100x cost savings & Orchestration of distributed training and inference pipelines
- Cost free Preemptible AI infrastructure hosting with savvy tooling and UIUX design
- Research & Engineer upon SOTA Multi-Modal, Multi Step Generative models with preference optimization to align generated AI influencers

Financial Data Engineer

Philadelphia, PA

Temple Fox School of Business

Jul 2023 - Jan 2025

- Increased 200,000 SOI data samples from 20 different BDCs to model SEC registered BDCs fiscal product categorization & valuation
- Eliminated need to post process data using OpenAIs gpt4o-mini, reducing data pipeline staging processes by 50%
- Prototyped and implemented staged Data models for ELT Data pipeline and Database for single source of truth mapping
- Containerized data extraction and data cleaning programs for future use and cross platform portability

Data Analytics Intern

Philadelphia, PA

Symetri USA

Jun 2024 - Sept 2024

- Abstracted away traditional data infrastructure using AlaaS paradigm, maintaining client data security and minimal technological overhead
- Migrated data infrastructure to Multi-Agent GenAI BIM Vector and Graph based Retrieval Augmented Generation POCs for client demonstration
- Landed partnerships with PennMedicine, several other Hospital Authorities and Port Authorities in PANJNY through built Compliance Frontend

Data Analytics Intern

Taipei, Taiwan

CPC Corporation Taiwan

Apr 2022 - Aug 2022

- Research qualitative analytics to CPC Taiwan Oil, whether having a new oil rig near the coast of Taoyuan damages its local ecology
- Replicated feature selection of expert lead data fields using Principle Component Analysis
- Structured Equational Models to conduct Conceptual causal analysis of latent regressor variables and observed response variables

AI Specialist Engineer

Taipei, Taiwan

Taiwan Tech Arena

May 2020 - Oct 2020

- Reduced fps stitching speed of 2x 3840X2160 pixels, 170 degree fisheye cameras from 6 seconds to 0.2 seconds on the Nvidia Jetson Xavier
- Reduced 0.2 to 0.01 fps using CUDA code integration of opencv 2.4 image blending into OpenCV 4.5 stitching during composePanorama
- EstimateTransform only once for real time composePanorama with multi-threading as camera is stationary but capture is 360 degrees
- 6x reduction in opencv cuda/cudnn binary compilation time by generating CPack binary deb files for single Opencv compilation (6 to 1 hours)
- Real time 3d video to 2d image projection homographies of player field positions
- Real time automated fisheye to panoramic view camera callibration to reconstruct 360 camera view to landscape view of sports field
- 6 main analysis features of mainly "camera panning", "zooming" and "slow motion"
- Ported Opencv C++ CUDA code between aarch64 tegra processors and x64-86 processors
- Patched opencv C++ customly optimized source code to opencv-python API via .so source file symlinks
- Automated video delivery and running of analysis from edge device to cloud instance via Apache Airflow dags

Research Experience

Publications

Pang, L., **Siu, T.**, Alazzawe, A., Kant, K., Latecki, L.J. (2026). Generalizable Detection of Student Engagement in On-line Learning Environments. In: Castrillón-Santana, M., et al. Computer Analysis of Images and Patterns. CAIP 2025. Lecture Notes in Computer Science, vol 15622. Springer, Cham. [🔗](#)

P.P Kumar, **Siu. T.**, S. Millwood, M.M Rogalska, K. Krishna and L.J. Latecki (2024). “Exploiting rPPG Signal Enhances Student Engagement Recognition in Online Video Education” *WACV*, 2025. [🔗](#)

Community Outreach

Skills

Work Focus	Multi-modal Generative AI, Computer Vision, Natural Language processing, Statistical Analysis, Data Engineering, Retrieval-Augmented Generation (RAG), Cloud System Administration
Programming	C, C++, CUDA, Rust, Java, JavaScript, TypeScript, Python, R, SQL, Bash ,Latex
Machine Learning	Pytorch, Tensorflow, Keras, Opencv, NLTK, Huggingface, Scikitlearn, OpenAI, Xgboost, Statsmodels, LangChain, LlamaIndex, Haystack, Hyperopt, Optuna
Scientific Computing	Pandas, Numpy, Scipy, Sympy
Data Visualization	Dash Plotly, Matplotlib, Streamlit, Paraview, VTK, D3.js, PowerBI
Infrastructure	Linux, Git, AWS, GCP, Docker, MySQL, MongoDB, PostgresSQL, Neo4j, ChromaDB, Apache Airflow, PowerAutomate, Docker, Docker-compose
WebDev	FastAPI, Django, Flask, Express.js, D3.js